

Ron Bibb
Independent Researcher
Lilburn, Georgia, USA
ronbibb@gmail.com

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Editors
General Relativity and Gravitation

Dear Editors,

Please consider my manuscript, “Surface-Energy Sign Classification for Timelike Kantowski–Sachs–Schwarzschild Junctions,” for publication in *General Relativity and Gravitation*. The manuscript identifies an invariant causal-sector split. Under the ordinary orientation, every compatible finite-rapidity junction from homogeneous Kantowski–Sachs geometry to the retained Schwarzschild $F > 0$ exterior requires negative Israel surface density. In the complementary $F < 0$ sector the ordering becomes sign-indefinite: the paper derives the exact positive-density interval and gives paired future-infalling timelike witnesses, verified in ingoing Eddington–Finkelstein coordinates, for which the same local bulk geometry and retained-side gluing admit opposite density signs when the longitudinal shell motion is reversed.

The significance of the result is that the Schwarzschild horizon becomes the invariant boundary between a universal negative-density exterior regime and a sign-indefinite interior regime. This explains geometrically why proposed black-hole-to-cosmology constructions placed inside the horizon are not subject to the exterior obstruction, without claiming to validate their dynamics or their generally spacelike transition layers. This combination of exact junction analysis, causal-sector classification, global branch control, and energy-condition analysis is well aligned with the scope of *General Relativity and Gravitation*.

The manuscript is original, is not under consideration elsewhere, and has been prepared independently without institutional or grant support. The manuscript source, figure-generation scripts, and verification suite are publicly available at github.com/RonBibb/junction-ec-paper.

Subject to the journal’s conflict-of-interest checks, potential referees with relevant expertise include João L. Rosa, Sebastian Carloni, S. H. Mazharimousavi, José M. M. Senovilla, and S. Khakshournia.

Sincerely,

Ron Bibb
Independent Researcher
Lilburn, Georgia, USA
ronbibb@gmail.com
ORCID: 0009-0004-1153-2464